TELCO CUSTOMER CHURN PREDICTION

End-to-End Machine Learning Project Report

Devixo Solutions — Week 3, AI/ML Internship, Task 03

This report describes the dataset, preparation steps, model comparison, tuned model results, limitations, and possible future improvements in simple language.

Dataset: Telco Customer Churn

Target: Churn (Yes / No)

Prepared from the project outputs shared during development

1. Dataset

The project uses the **Telco Customer Churn** dataset from Kaggle. It contains customer account and service information and a label showing whether each customer left the service.

Source: https://www.kaggle.com/datasets/blastchar/telco-customer-churn
Raw size: 7,043 rows and 21 columns.
Target column: `Churn` (Yes or No). After one-hot encoding, the target may appear as `Churn_Yes`.
Example features: tenure, contract type, internet service, monthly charges, total charges, and payment method.

2. Workflow

1. Load and inspect	Load the CSV and check its first rows, shape, column types, and summary statistics.
2. Explore the data	Create charts for churn counts, churn by contract, churn by internet service, monthly charges by churn, and numeric correlations.
3. Clean missing values	Convert TotalCharges to numeric and fill missing numeric values with medians and categorical values with modes.
4. Remove duplicates	Check duplicate rows and remove them if any are present.
5. Engineer features	Drop customerID, create TenureGroup, and create ChargesPerTenureMonth using monthly charges and tenure.
6. Encode categories	Use one-hot encoding to convert categorical columns into numeric features.
7. Select features	Use SelectKBest with the ANOVA F-test to select up to 15 features.
8. Split and scale	Create training and test sets, then fit StandardScaler on the training data and transform the data.
9. Train four models	Train Decision Tree, Random Forest, SVC, and Gradient Boosting.
10. Tune the model	Use RandomizedSearchCV to test six Random Forest settings with 3-fold cross-validation.
11. Cross-validation	A separate 5-fold cross_val_score step was requested. Copy its actual five scores and mean from the final notebook output before submission.
12. Feature importance	Plot the tuned Random Forest feature_importances_ in descending order.
13. Save and test	Save the model with Joblib and Pickle, save the scaler and column names, reload them, and make a prediction.

3. Model Selection

The tuned Random Forest was selected as the final model because its reported churn-class F1-score (0.6161) was the strongest among the models listed here, and its churn recall (0.7166) was higher than the original Random Forest (0.47). This means it identified a larger share of customers who actually churned. Gradient Boosting had the highest reported accuracy (0.7928), but its churn recall was 0.50. For churn prediction, missing customers who may leave can matter, so accuracy alone is not enough to choose a model.

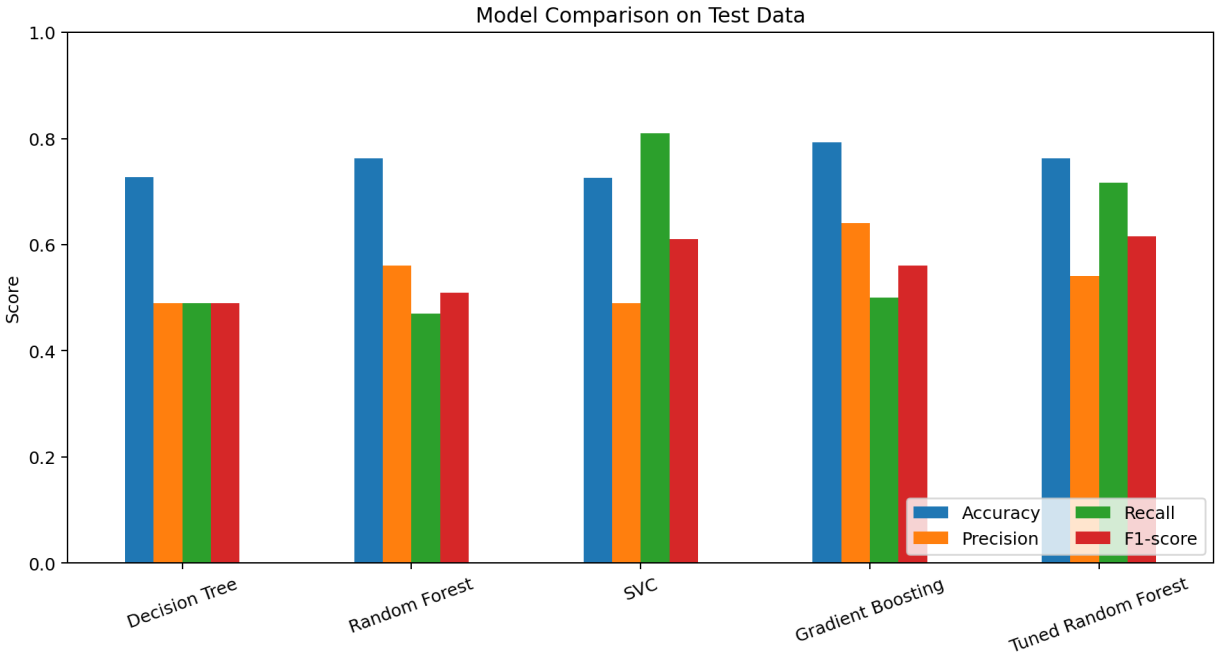
4. Results

The table below summarizes the metrics reported during development. Precision, recall, and F1-score refer to the positive class (Churn = Yes). Baseline metrics are rounded as reported. ROC-AUC values for the baseline models were not available in the outputs shared for this report.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Decision Tree	0.7275	0.4900	0.4900	0.4900	Not recorded
Random Forest	0.7622	0.5600	0.4700	0.5100	Not recorded

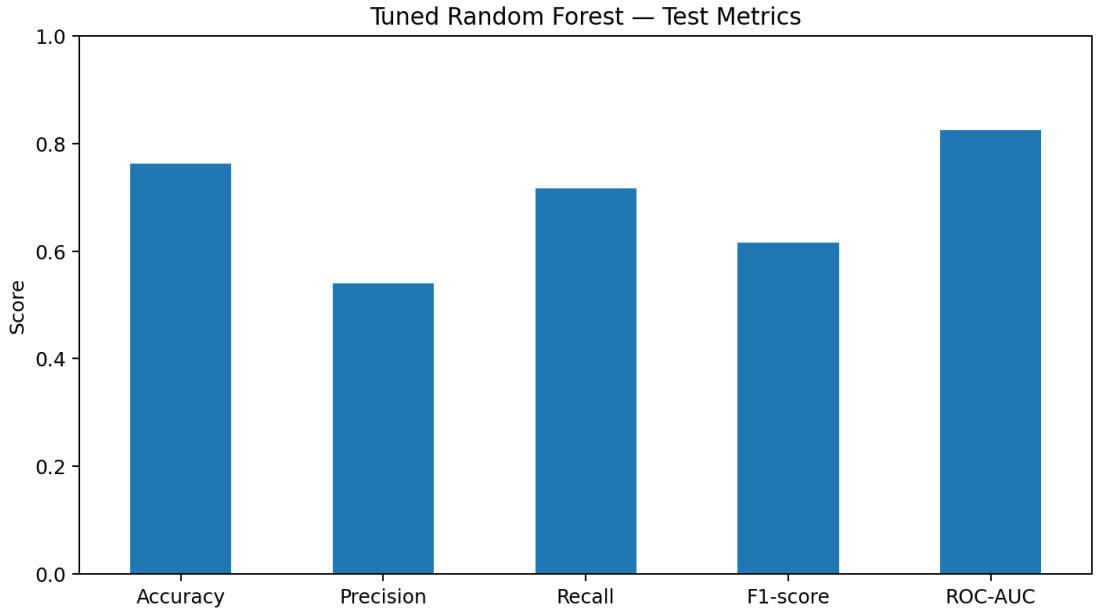
Model	Accuracy	Precision	Recall	F1	ROC-AUC
SVC	0.7260	0.4900	0.8100	0.6100	Not recorded
Gradient Boosting	0.7928	0.6400	0.5000	0.5600	Not recorded
Tuned Random Forest	0.7630	0.5403	0.7166	0.6161	0.8259

Model comparison chart



Tuned Random Forest Results

Best parameters: n_estimators = 50; max_depth = 10; min_samples_split = 2; min_samples_leaf = 1.
Best 3-fold tuning CV F1-score: 0.6113.
Test metrics: accuracy = 0.7630; precision = 0.5403; recall = 0.7166; F1-score = 0.6161; ROC-AUC = 0.8259.



Note: the 0.6113 cross-validation score is from the 3-fold RandomizedSearchCV tuning run. It is not the result of the separate 5-fold cross-validation cell. Add the five scores and their average from that cell once available.

5. Limitations

• The dataset is imbalanced: non-churn customers are more common than churn customers.

• The project uses one public dataset, so results may not generalize to every telecom provider.

• Tuning was limited to six sampled parameter settings and 3-fold CV to keep runtime manageable.

• Feature selection and some preprocessing were performed before the train/test split in the current workflow. This can cause data leakage and make scores optimistic.

6. Future Improvements

• Use newer data and test the model on another time period or provider.

• Move imputation, feature selection, and scaling into a scikit-learn pipeline and evaluate without data leakage.

• Try more hyperparameter settings and compare with XGBoost or other boosting methods.

• Study precision-recall curves and choose a prediction threshold based on the business cost of missed churners.

• Deploy a simple Streamlit prediction interface.

Conclusion

This project completes the main steps of an end-to-end churn prediction workflow: data preparation, feature creation and selection, model comparison, tuning, feature importance, saving, and reloading. The tuned Random Forest is a practical candidate because it improves churn recall over the baseline Random Forest. Before final submission, add the exact 5-fold cross-validation output and consider re-running preprocessing and feature selection inside a pipeline to avoid data leakage.

Author: Add your name and internship details before submission.